



User Guide

A Web-Based Real-Time Sign Language Detection & Learning Platform

Version 1.0

Document Information

Product Name: SUHWA

Version: 1.0

Platform Type: Web-Based Application

Primary Technology: AI-Based Sign Language Detection

Target Users: Students, Educators, Deaf & Hard-of-Hearing Community

Table of Contents

1. Introduction
 2. Purpose of the System
 3. System Requirements
 4. Accessing the Platform
 5. User Account Management
 6. Dashboard Overview
 7. Detect Module
 8. Learn Module
 9. AI Tutor Module
 10. Quiz & Assessment
 11. Video Upload Feature
 12. Progress & Analytics
 13. Data Privacy & Security
 14. Troubleshooting Guide
 15. Support & Contact Information
-

1. Introduction

SUHWA is a web-based real-time sign language detection and learning platform designed to support inclusive communication. The system integrates artificial intelligence and computer vision to recognize hand gestures and convert them into readable text.

In addition to detection, SUHWA provides structured learning resources and an AI-driven tutor to enhance user engagement and knowledge retention.

2. Purpose of the System

The primary objectives of SUHWA are:

- To provide real-time sign language recognition
 - To offer structured sign language learning modules
 - To enable interactive AI-based tutoring
 - To support inclusive communication environments
 - To provide academic and educational support
-

3. System Requirements

Hardware Requirements

- Desktop or Laptop Computer
- Functional Webcam (Built-in or External)
- Minimum 4GB RAM
- Stable Internet Connection

Software Requirements

- Google Chrome (Recommended)
 - Microsoft Edge or Mozilla Firefox
 - Camera permission enabled in browser
-

4. Accessing the Platform

1. Open the SUHWA website in your browser.

2. Click on “Sign In” or “Register”.
3. Allow camera permissions when prompted.
4. Access the dashboard after login.

5. User Account Management

Registration Process

Users must provide:

- Full Name
- Email Address
- Password

After registration, users can log in using their credentials.

Login Process

- Enter registered email
- Enter password
- Click “Sign In”

Upon successful login, users are redirected to the dashboard.

6. Dashboard Overview

The dashboard serves as the central navigation hub and includes the following sections:

- Home
- Detect
- Learn
- AI Tutor
- Quiz
- Upload Video
- Progress
- Documentation

- [FAQ](#)
- [Contact](#)

The dashboard also displays:

- Learning progress summary
 - Accuracy statistics
 - Recent activities
 - Recommended lessons
-

7. Detect Module

Purpose

The Detect Module allows real-time recognition of sign language using the webcam.

How to Use

1. Click “Detect” .
2. Grant camera access.
3. Position your hand clearly in front of the camera.
4. Perform a sign gesture.
5. The system will display:
 - Recognized text output
 - Prediction confidence percentage
 - Accuracy feedback

Best Practices for Accuracy

- Ensure proper lighting
 - Keep background simple
 - Show only one hand clearly
 - Maintain steady hand position
-

8. Learn Module

The Learn Module provides structured educational content.

8.1 Alphabet Learning (A – Z)

Each alphabet includes:

- Visual demonstration
- Hand gesture description
- Practice option
- Camera verification
- Instant feedback

8.2 Numbers (0 – 10)

Users can:

- View number signs
- Practice with guided instructions
- Attempt timed exercises

8.3 Basic Words (20 Essential Words)

Includes common conversational signs such as:

- Hello
- Thank You
- Please
- Sorry
- Yes
- No
- Help
- Love
- Water
- Study

Each word contains:

- Gesture explanation
- Visual guidance

- Practice functionality
-

9. AI Tutor Module

The AI Tutor is an interactive learning assistant.

Features

- Requests users to perform specific signs
- Provides immediate evaluation
- Offers improvement suggestions
- Simulates basic sign-based conversations

Example Interaction

AI: “Show the sign for ‘Thank You’ .”

User performs gesture.

System evaluates and provides feedback.

The AI Tutor also tracks user performance to personalize future sessions.

10. Quiz & Assessment

The Quiz Module evaluates user understanding.

Quiz Types

1. Multiple Choice Questions
2. Camera-Based Gesture Verification

Scoring System

- Beginner: 0 - 50%
- Intermediate: 51 - 80%
- Advanced: 81 - 100%

Users can review incorrect answers and reattempt quizzes.

11. Video Upload Feature

This feature allows users to upload pre-recorded sign language videos.

Steps

1. Click “Upload Video” .
 2. Select a supported video file (MP4, MOV, AVI).
 3. The system analyzes video frames.
 4. Detected output is displayed.
-

12. Progress & Analytics

Users can track:

- Total signs practiced
- Accuracy percentage
- Quiz scores
- Performance trends

Visual graphs and analytics help monitor improvement.

13. Data Privacy & Security

SUHWA prioritizes user privacy.

- Camera data is processed temporarily.
 - No permanent storage of live camera footage.
 - User account information is securely stored.
 - Passwords are encrypted.
-

14. Troubleshooting Guide

Camera Not Working

- Enable camera permissions
- Refresh the page
- Restart browser

Low Detection Accuracy

- Improve lighting conditions
- Remove background distractions

- Maintain correct hand positioning

System Lag

- Close unnecessary applications
- Use recommended browser
- Check internet speed

15. Support & Contact Information

For technical assistance or inquiries:

Email: yellowversecenter@gmail.com

Website: www.suhwa.com

Conclusion

SUHWA is designed to make sign language learning accessible, interactive, and technologically advanced. Through AI-powered detection and structured learning modules, the platform promotes inclusive communication and educational empowerment.

We encourage users to practice regularly and explore all modules for optimal learning outcomes.